

TMS, DMS and CMS Usage Guide for Falcon BMS 4.38.1

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Contents

List of HOTAS Tables	ii
0.1 TMS in Air-to-Air	1
0.1.1 TMS in SAM (Situation Awareness Mode)	2
0.1.2 TMS in STT (Single Target Track)	2
0.1.3 TMS in TWS (Track-While-Scan)	3
0.1.4 TMS in ACM (Air Combat Maneuvering) modes	4
0.1.5 TMS and IFF Interrogation	5
0.1.5.1 SCAN Interrogation Mode (TMS Left Short Press)	6
0.1.5.2 LOS Interrogation Mode (TMS Left Long Press)	6
0.1.5.3 TMS IFF Interrogation Usage Table	6
0.1.5.4 Operational Constraints and Notes	6
0.1.5.5 AIFF Availability and Block Differences	6
0.1.6 TMS with Targeting Pod in Air-to-Air Mode	6

List of HOTAS Tables

Perfeito! Então é progresso parcial na S5.

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0.1 TMS in Air-to-Air

The Target Management Switch (TMS) in air-to-air operations provides hands-on control for target designation, track management, and sensor cueing during weapons employment. The pilot engaged in beyond-visual-range (BVR) intercepts or within-visual-range (WVR) maneuvering cannot divert attention to MFD interaction or remove hands from the flight stick. TMS commands deliver immediate control of radar tracking modes, target prioritization, and identification functions without interrupting aircraft control or visual scan.

As established in Section ?? and detailed in Section ??, A-A master mode restricts Sensor of Interest (SOI) designation to MFD formats only—specifically, the Fire Control Radar (FCR), Horizontal Situation Display (HSD), and Targeting Pod (TGP). Consequently, TMS commands in A-A are restricted to FCR, HSD, and TGP formats; the HUD, although providing essential cueing (Section ??), does not receive TMS inputs in this master mode. The Helmet Mounted Cueing System (HMCS) provides independent off-boresight target cueing capability and AIM-9 seeker slaving, but TMS inputs route through the SOI-designated MFD sensor, not through HMCS or HUD symbology.

TMS behavior in air-to-air varies by sensor mode and tracking state. The primary tactical employment of TMS occurs through FCR tracking modes, which manage the complete engagement cycle from search to weapons delivery. Beyond FCR tracking, TMS Left commands IFF (Identification Friend or Foe) interrogation, critical for Rules of Engagement (ROE) compliance before BVR weapons release, and TMS with TGP as SOI manages visual identification and backup tracking independent of radar state. The FCR provides four primary tracking modes, each optimized for specific tactical contexts:

- **Situation Awareness Mode (SAM)** (Dash-34 § 2.3.1.5.9) tracks one or two targets while maintaining search coverage, balancing situational awareness with engagement capability.
- **Single Target Track (STT)** (Dash-34 § 2.3.1.5.8) dedicates the radar to a single target with maximum accuracy, but continuously radiates and reveals the F-16's position to threat receivers.
- **Track-While-Scan (TWS)** (Dash-34 § 2.3.1.5.7) automatically builds and maintains track files on up to ten targets while continuing to search, optimizing multi-bogey BVR engagements.
- **Air Combat Maneuvering (ACM)** (Dash-34 § 2.3.1.5.10–Dash-34 § 2.3.1.5.12) modes provide four automated scan patterns (BORE, 10×60, 30×20, SLEW) optimized for close-range visual acquisition with rapid lock and break-lock cycles.

TMS commands manage not only tactical functions within each mode, but also transitions between modes. The pilot transitions between SAM, STT, TWS, and search modes (RWS) using TMS inputs, with the resulting mode depending on the current radar state and targeting context. The same physical TMS input produces different results based on whether a target is bugged, whether the radar is in single-target or multi-target tracking, and whether the pilot is prosecuting a primary threat or re-establishing search coverage. This state-dependency requires the pilot to maintain awareness of current radar mode and tracking status to predict TMS behavior accurately.

The following subsections detail TMS behavior in each air-to-air tracking context. [Section 0.1.1](#) through [Section 0.1.4](#) examine FCR modes (SAM, STT, TWS, ACM), [Section 0.1.5](#) covers IFF interrogation, and [Section 0.1.6](#) addresses TGP air-to-air tracking. Each subsection includes detailed HOTAS tables documenting specific TMS inputs, effects, Dash-34 cross-references, and recommended training missions for hands-on practice.

Note: TMS with HSD for datalink target management (PDLT designation) is covered in [Section ??](#). While PDLT is an air-to-air function, HSD is primarily a situational awareness display rather than a direct targeting sensor, and its TMS procedures differ from FCR, TGP, and IFF operations documented in this section.

0.1.1 TMS in SAM (Situation Awareness Mode)

Situation Awareness Mode (SAM) tracks one or two targets (ST SAM or DT SAM) while maintaining search coverage. TMS commands in SAM manage track transitions and target prioritization.

TMS Up in SAM:

- **On bugged target:** Transition SAM → STT (Single Target Track). Search coverage ceases; FCR dedicates all resources to tracking the bugged target.
- **On search target (DT SAM):** Designate second target as secondary SAM target, entering Dual Target SAM (DT SAM). Radar tracks both targets while searching.

TMS Down in SAM (Return-to-Search):

- **ST SAM:** Drops bugged target, returns to RWS search mode. Target becomes untracked search symbol.
- **DT SAM:** Drops secondary target, returns to ST SAM with primary target still bugged.

TMS Right in SAM (short press <1s):

- **DT SAM:** Toggles bug between primary and secondary targets. Allows pilot to prioritize which target is primary without re-designation.

TMS Right in SAM (held >1s): Transitions SAM → TWS. Bugged target becomes TWS primary target. Radar begins building track files on other search targets.

Operational context: SAM is ideal for Beyond Visual Range (BVR) engagements where the pilot needs track data on one or two targets while maintaining search awareness. TMS Up/Down provides rapid transitions between search, SAM, and STT as the tactical picture evolves.

See [Table ??](#) for detailed SAM TMS actions.

0.1.2 TMS in STT (Single Target Track)

Single Target Track (STT) dedicates the FCR to tracking a single airborne target with maximum accuracy. All search processing ceases. TMS commands in STT are limited to breaking lock and returning to previous modes.

TMS Down in STT (Return-to-Search):

- **First press:** STT → SAM (or DT SAM if secondary target was being extrapolated). STT target becomes SAM primary target.
- **Second press:** SAM → RWS. Bugged target becomes untracked search symbol.

TMS Up, TMS Right, TMS Left in STT:

- **ACM STT:** TMS Up, TMS Right, or CURSOR slew do NOT break lock. This allows pilot to slew scan volume or step targets without losing track (ACM-specific behavior).
- **Non-ACM STT:** TMS Up/Right/Left have no effect. Only TMS Down breaks lock.

Operational note: STT reveals the F-16's position to any target with RWR (Radar Warning Receiver) due to continuous radar emissions. Use STT only when weapon parameters are achieved and giving away position is acceptable. For covert BVR engagements, SAM or TWS is preferable.

See Table ?? for STT TMS actions.

0.1.3 TMS in TWS (Track-While-Scan)

Track-While-Scan (TWS) automatically builds and maintains track files on up to 10 targets while continuing to search. TMS commands manage target prioritization (bugging) and transitions to STT for weapons employment.

TMS Up in TWS:

- **Cursor over track file:** Bug that target (make it primary/TOI). If already bugged, transition TWS → STT on bugged target.
- **Cursor over search target:** Designate search target, build track file, and bug it.

TMS Down in TWS (Return-to-Search):

- **First press:** Drop bug (if bugged target exists). Radar continues TWS with all track files.
- **Second press:** Dump all track files, return to RWS search mode.
- **Third press:** (from RWS after two presses) Radar rebuilds tracks automatically.

TMS Right in TWS (short press <1s): Steps bug to next highest-priority track file. Priority is determined by range and track file order.

TMS Right in TWS (held >1s):

- **With bugged target:** TWS → DT SAM. Bugged target becomes SAM primary; highest-priority track file becomes SAM secondary.
- **Without bugged target:** TWS → RWS (last exited CRM search mode).

TMS Up (held >1s) in TWS: Initiates spotlight search (same as RWS). Spotlight scan centered on ACQ cursor; track updates on TOI and AIM-120 in-flight targets occur only if spotlight volume includes them.

Operational context: TWS excels in multi-target environments (BARCAP, HAVCAP, large force engagements). Pilot can maintain SA on 8-10 threats while bugging and prosecuting the highest-priority target. TMS Right stepping allows rapid target prioritization without MFD interaction.

See Table ?? for TWS TMS actions.

0.1.4 TMS in ACM (Air Combat Maneuvering) modes

Air Combat Maneuvering (ACM) modes provide automated close-range target acquisition (BORE, 10×60, 30×20, SLEW). TMS commands manage mode transitions, target rejection, and HMCS integration.

TMS Up in ACM

ACM BORE with HMCS (TMS Up held):

- **Hold phase (TMS Up depressed):**
 - FCR scan slaves to HMCS aiming cross position.
 - Radar in NO RAD state (non-radiating) — pilot aims scan volume with head movement.
 - ACQ cursor tracks HMCS aiming cross.
 - No minimum hold duration — pilot controls timing.
- **Release phase (TMS Up released):**
 - FCR radiates at HMCS-designated position.
 - Auto-acquisition attempts on target under aiming cross.
 - If successful: Enters STT, solid TD box displayed on HUD.
 - If unsuccessful: Remains in ACM BORE, radiating.
- **Tactical use:**
 - High off-boresight (HOBS) acquisition: Lock target 60–90° off nose without pointing aircraft.
 - Visual merge: Identify bandit visually, slave radar via HMCS, acquire without nose alignment (critical in scissors/rolling fights).
 - Target sorting: Acquire specific bandit in multi-bogey merge without broad scan hitting wrong target.

ACM BORE without HMCS (TMS Up held):

- Holding TMS Up inhibits auto-acquisition until released.
- Allows pilot to position boresight cross over desired target in clustered group before commanding acquisition (manual target sorting).
- No HMCS slaving — radar remains boresight-aligned.

ACM STT:

- TMS Up, TMS Right, or CURSOR slew does NOT break lock (ACM-specific behavior).
- Allows pilot to reposition scan or step targets without losing track.

TMS Down in ACM

Return-to-Search (any radiating ACM mode):

- First press: Drops radar track (if any), commands 30×20 NO RAD.
- Second press: Switches to 10×60 radiating.
- Third press: Returns to 30×20 NO RAD.

HUD as SOI with FCR slaved to HMCS:

- TMS Down commands ACM BORE NO RAD.

TMS Right in ACM

- Cycles through ACM scan patterns (implementation varies by mode).
- Typical sequence: $30 \times 20 \rightarrow \text{SLEW} \rightarrow 10 \times 60 \rightarrow \text{BORE}$.

Operational context

ACM modes automate acquisition in visual merge/dogfight. TMS Up/Down provide rapid lock/break-lock cycles essential in turning fights. HMCS integration enables HOBS acquisition without pointing aircraft nose.

Reference: ACM modes details in Dash-34 § 2.3.1.5.10. HMCS integration in Dash-34 § 2.5.

See Table ?? for ACM TMS actions.

0.1.5 TMS and IFF Interrogation

Identification Friend or Foe (IFF) interrogation is an air-to-air identification function commanded by TMS Left, enabling the pilot to distinguish friendly aircraft from unknowns and bogeys without removing hands from the flight stick. The interrogation mode executed depends on press duration: a short press activates SCAN interrogation for broad area coverage, while a long press activates Line of Sight (LOS) interrogation for precise single-target queries. Both modes display interrogation responses on the Fire Control Radar (FCR) MFD with symbology overlaid on radar contacts, enabling immediate threat discrimination during multi-target engagements without requiring MFD interaction or datalink correlation (see [Section 0.1.5.1](#) and [Section 0.1.5.2](#) for details).

IFF interrogation is operationally critical in beyond-visual-range (BVR) engagements, where visual identification is impossible and Rules of Engagement (ROE) require positive identification before weapons employment to prevent fratricide. The interrogation process begins when the interrogating aircraft transmits a coded query to the target aircraft, whose transponder validates the query and transmits a response back to the interrogator. The FCR then displays identification symbology based on the response received: a green circle indicates a friendly aircraft that provided a correct response, while a yellow square identifies a bogey or unknown contact that provided an incomplete response or failed to respond.

Not all F-16 variants are equipped with IFF interrogation capability, as the AIFF (Advanced Interrogator Friend or Foe) system is not universally installed across all F-16 blocks and configurations. When available, TMS Left IFF interrogation requires the AIFF system to be powered and configured via the Up Front Controller (UFC) or backup IFF panel controls. Pilots must verify AIFF availability via cockpit panel configuration before flight; see [Section 0.1.5.5](#) for AIFF identification procedures and system limitations. For complete IFF system architecture, operational modes, and control panel operation, see Dash-34 § 2.2.4.

0.1.5.1 SCAN Interrogation Mode (TMS Left Short Press)**0.1.5.2 LOS Interrogation Mode (TMS Left Long Press)****0.1.5.3 TMS IFF Interrogation Usage Table****0.1.5.4 Operational Constraints and Notes****0.1.5.5 AIFF Availability and Block Differences****0.1.6 TMS with Targeting Pod in Air-to-Air Mode**

The Targeting Pod (TGP) operates in Air-to-Air mode for visual/IR target identification and tracking independent of FCR. TMS commands with TGP as SOI manage POINT track designation and track rejection.

TMS Up (TGP SOI, A-A mode)

- Commands POINT track on center of TGP field-of-view (crosshair position).
- TGP locks onto visual/IR target at designated position.
- After track command, TGP LOS becomes independent of FCR LOS.
- HUD displays:
 - Dotted 50-mr TD box (TGP track position).
 - Solid 50-mr TD box (FCR track position) if FCR also tracking.
 - Dotted TLL (Target Locator Line) with “T” + angle if TGP track outside HUD FOV.

TMS Down (TGP SOI, A-A mode)

- Breaks TGP POINT track.
- Returns TGP to slave mode — TGP re-slaves to FCR LOS.
- Dotted TD box/TLL removed from HUD.

Operational context

TGP A-A mode provides:

1. **Visual ID (VID):** Identify bandits before BVR shot (ROE compliance).
2. **Dual tracking:** FCR tracks primary threat, TGP tracks secondary (2v1 scenarios).
3. **Backup tracking:** If FCR loses lock (chaff/jamming), TGP maintains visual track.
4. **High-aspect targets:** TGP may track targets FCR cannot resolve (low RCS, beam aspect).

Limitations

- TGP A-A track is passive (no ranging) — uses triangulation for range estimate.
- Track quality degrades at long range (>10 NM) or high closure rates.
- TGP slew rate slower than FCR — not suitable for high-G dogfight tracking.

Procedure (VID example)

1. FCR in RWS/SAM with unknown contact.
2. Select A-A master mode, TGP powered.
3. DMS Down to make TGP SOI.
4. Slew TGP crosshair over bandit (TGP initially slaved to FCR LOS).
5. TMS Up — TGP commands POINT track, dotted TD box appears on HUD.

6. Visually identify via TGP video (MFD TGP page).

7. TMS Down — break TGP track after ID complete.

Reference: TGP A-A mode in Dash-34 § 2.6.1.7.5. SRM procedures with TGP in Dash-34 § 4.3.2.4.3.12.

See Table ?? for TGP A-A TMS actions.

[PLACEHOLDER: The following tables will be developed with full hotastable format:]